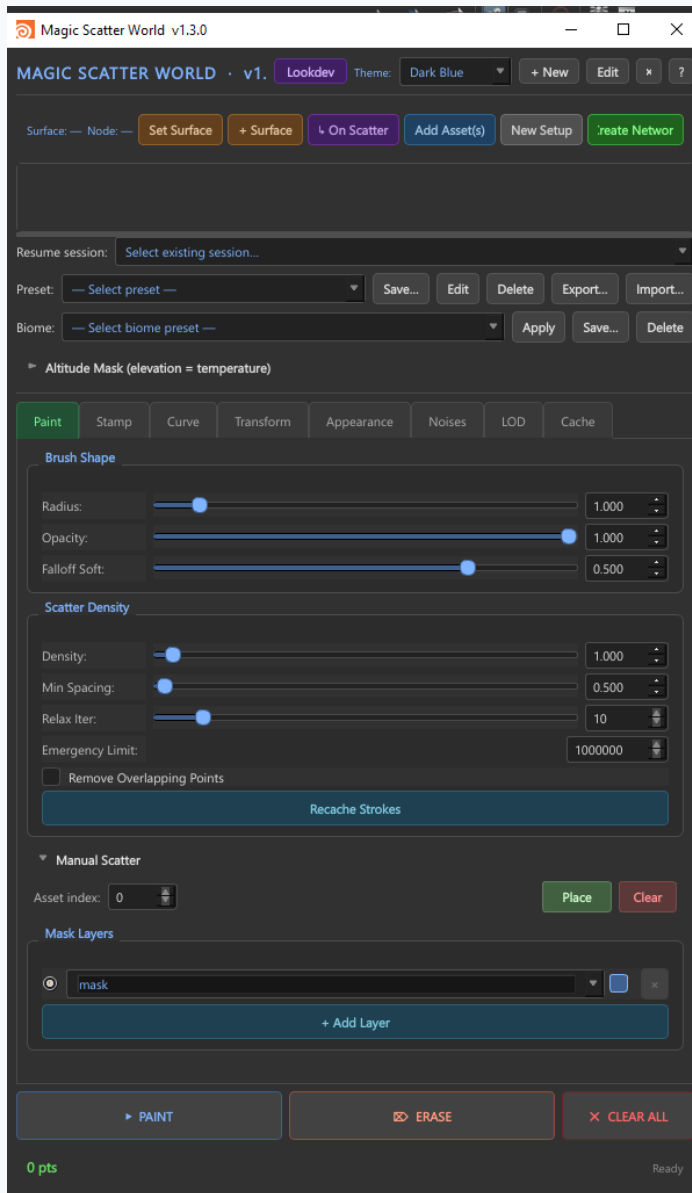


Magic Scatter World

v1.3.0 · Complete Tutorial Guide

Brush-based scatter painting for Houdini · Arnold · Redshift · Karma / USD

Paint Tab · Brush painting · density control · mask layers



Brush Shape

Radius — world-space brush size in the viewport

Opacity — stroke strength; lower values build up gradually

Falloff Soft — 0 = hard circular edge · 1 = smooth fade

Scatter Density

Density — instances placed per unit area

Min Spacing — minimum distance enforced between any two instances

Relax Iter — relaxation passes for even, natural distribution

Emergency Limit — hard point count cap (prevents runaway cooks)

Remove Overlapping Points — post-process pass to fix remaining overlaps

Recache Strokes — force full re-evaluation of all painted strokes

Manual Scatter

Asset index — choose which asset to hand-place

Place — drop one instance at cursor · Clear — remove manual instances

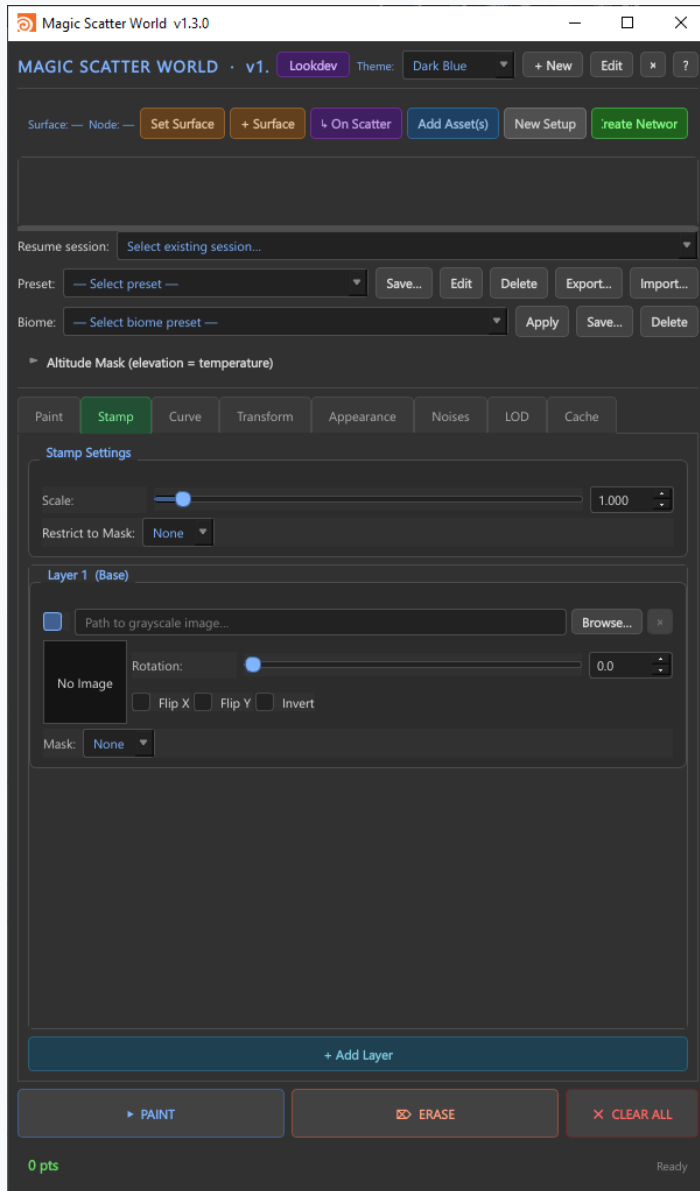
Mask Layers

Painted or procedural density masks stacked in order

Active layer receives your brush strokes

+ Add Layer — build complex density maps with multiple layers

Stamp Tab · Paint with custom grayscale image brushes



Stamp Settings

Scale — stamp size relative to the brush radius

Restrict to Mask — limit stamp painting to a specific painted region

Layer 1 (Base)

Path to grayscale image — white = full density · black = nothing

Supports .png · .jpg · .tif · .exr

Rotation — rotate the stamp image per stroke

Flip X / Flip Y — mirror horizontally or vertically

Invert — swap black and white; creates clearings or holes

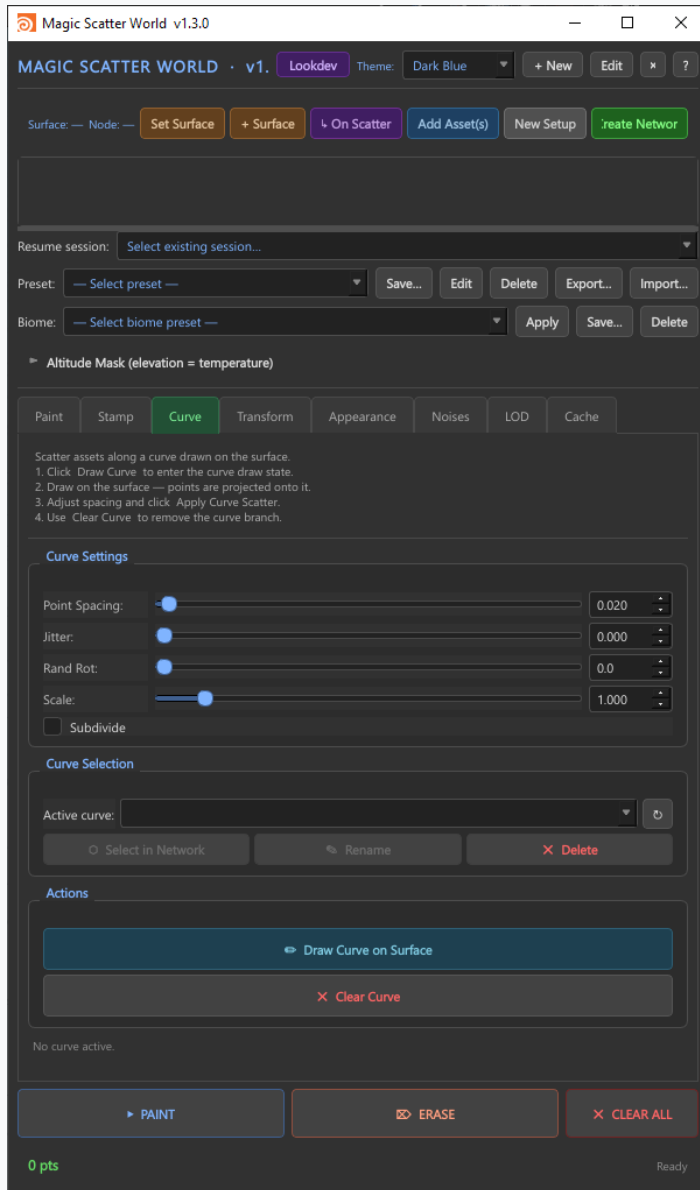
Mask — secondary mask controlling where this layer is allowed

Multiple Layers

+ Add Layer to stack stamp images on top of each other

Combine a cluster silhouette + noise texture in one brush stroke

Curve Tab · Scatter along hand-drawn curves projected onto the surface



Workflow

1. Click Draw Curve on Surface to enter draw mode
2. Click on geometry in the viewport to place control points
3. Curve is automatically projected onto the surface
4. Adjust spacing / variation · use Clear Curve to start over

Curve Settings

Point Spacing — distance between instances along the curve

Jitter — random perpendicular offset for organic rows

Rand Rot — random per-instance Y rotation in degrees

Scale — uniform scale multiplier for curve-scattered instances

Subdivide — smoother distribution on tightly curved paths

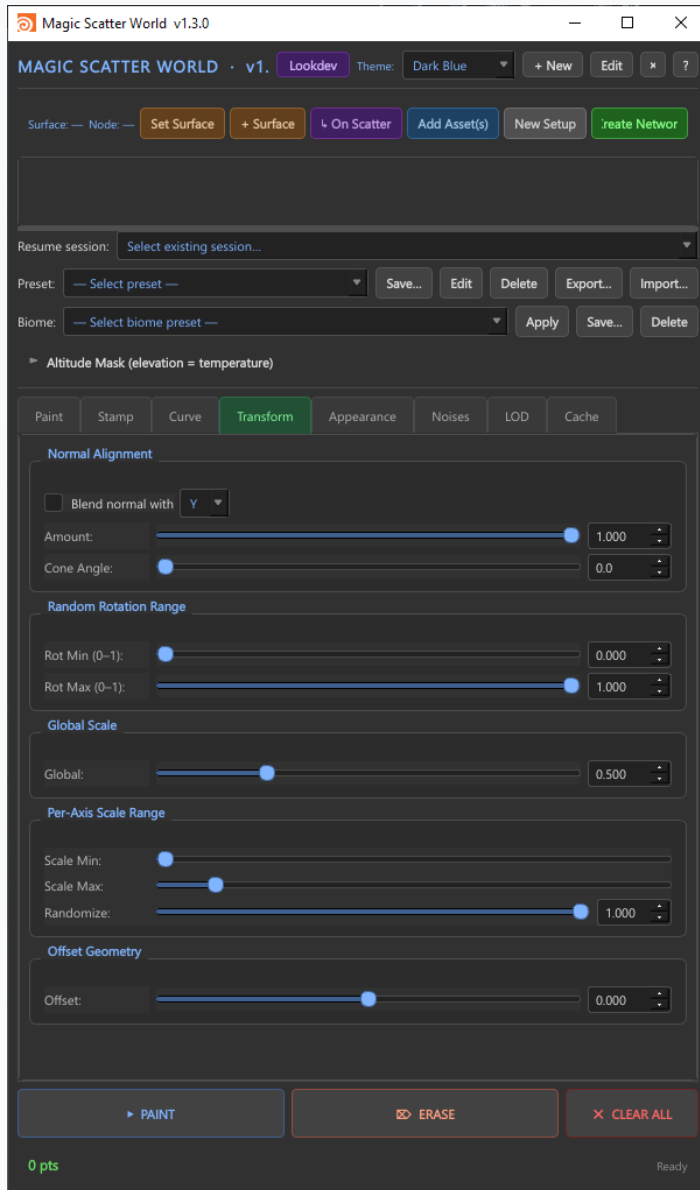
Curve Selection

Active curve dropdown — switch between multiple drawn curves

Select in Network — jump to curve node in the Network Editor

Rename / Delete — manage your curve library

Transform Tab · Orientation, rotation, and scale for every instance



Normal Alignment

Amount — 1.0 = aligned to surface normal · 0.0 = aligned to world axis

Blend normal with — world axis to tilt toward as Amount decreases (Y = up)

Cone Angle — random tilt within a cone; simulates organic lean and wobble

Random Rotation Range

Rot Min / Max (0–1) — range of Y-axis spin per instance

0 → 0° · 1 → 360°

Global Scale

Uniform scale multiplier applied to every instance

Per-Axis Scale Range

Scale Min / Max — random scale range; each instance gets a unique value

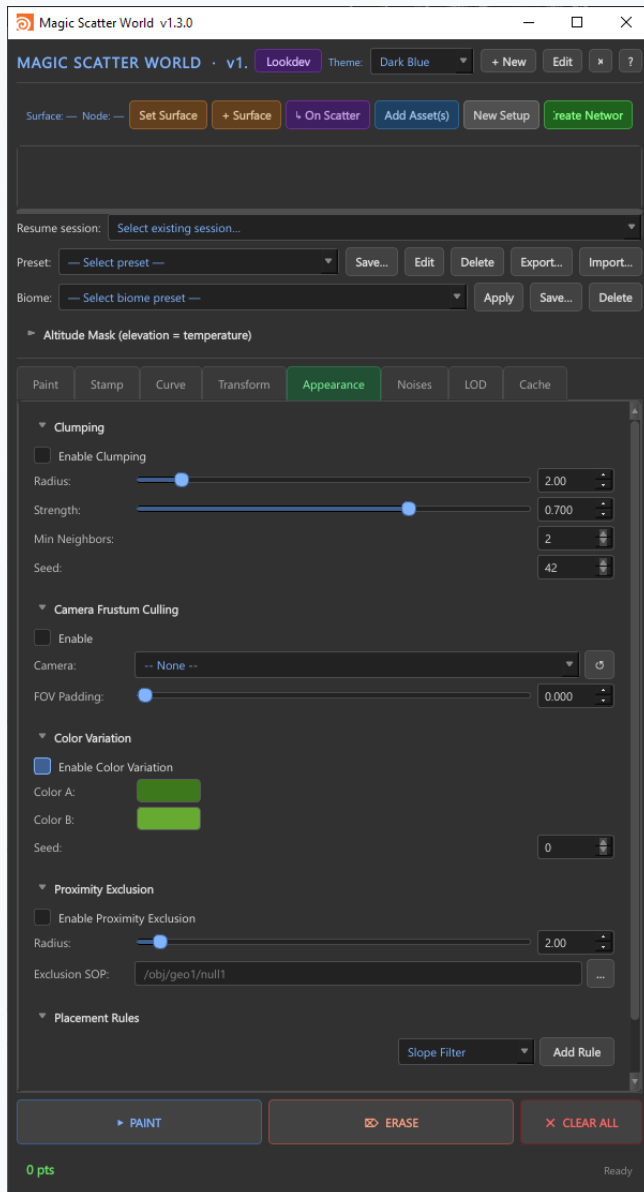
Randomize — dial back variation while keeping Min / Max set for later

Offset Geometry

Moves each instance along its surface normal

Use to lift assets above dense ground cover planes

Appearance Tab · Clumping · color · proximity exclusion · placement rules



Clumping

Groups nearby instances into natural colonies

Radius — neighbor search distance

Strength — how strongly points pull toward cluster centers

Min Neighbors — cull isolated stragglers below this count

Seed — change to get a different cluster pattern

Camera Frustum Culling

Removes off-screen instances to save memory and cook time

FOV Padding — expands frustum to prevent edge-of-screen popping

Color Variation

Per-instance color blend between Color A and Color B

Drive in shader for leaf tint, bark color, rock hue variation

Proximity Exclusion

Deletes instances within Radius of a reference SOP

Use to clear scatter away from rocks, walls, and props

Placement Rules

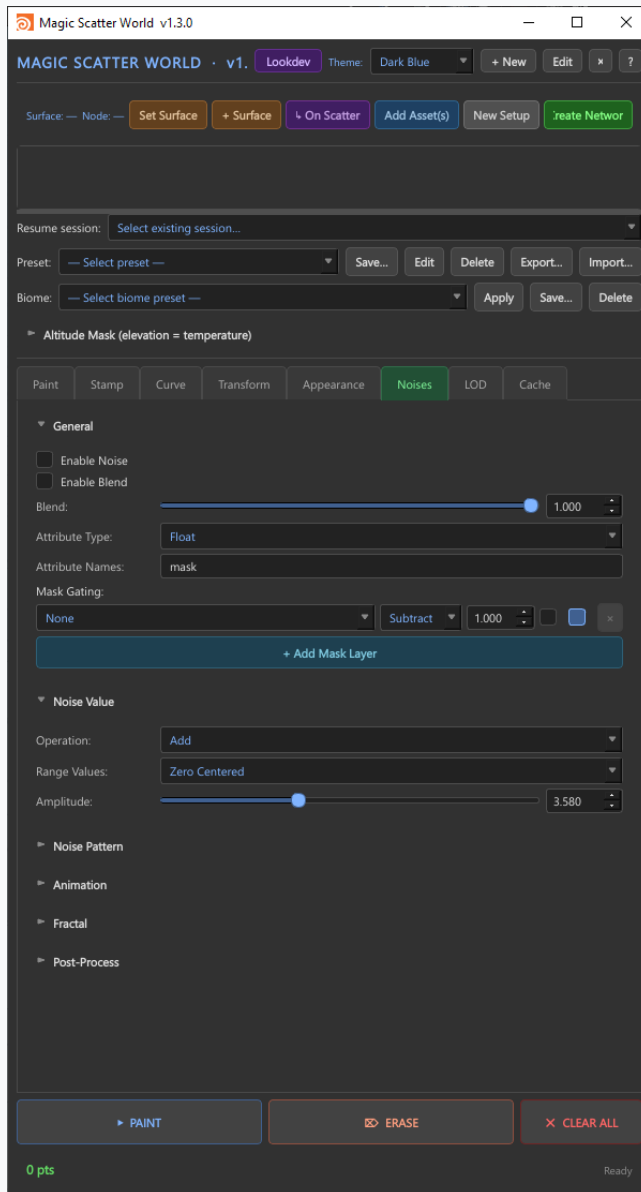
Slope Filter — remove instances beyond a max surface angle

Altitude Filter — keep instances within a Y elevation range

Noise Threshold — patchy distribution without manual painting

Distance from Path — clear zone around a curve or geometry

Noises Tab · Procedural noise field to modulate scatter density



General

Enable Noise — activates the procedural density modulation field

Enable Blend — softer mix into painted mask vs hard multiply

Blend — 1.0 = noise fully drives density · 0.0 = painted mask only

Attribute Names — point attribute written by noise (default: mask)

Mask Gating — secondary gate with blend mode and threshold value

Noise Value

Operation — Add / Multiply / Replace the existing painted mask

Range Values — Zero Centered ($\pm$ amp) or Zero to One (positive only)

Amplitude — strength of the sparse / dense contrast

Advanced (collapsible sections)

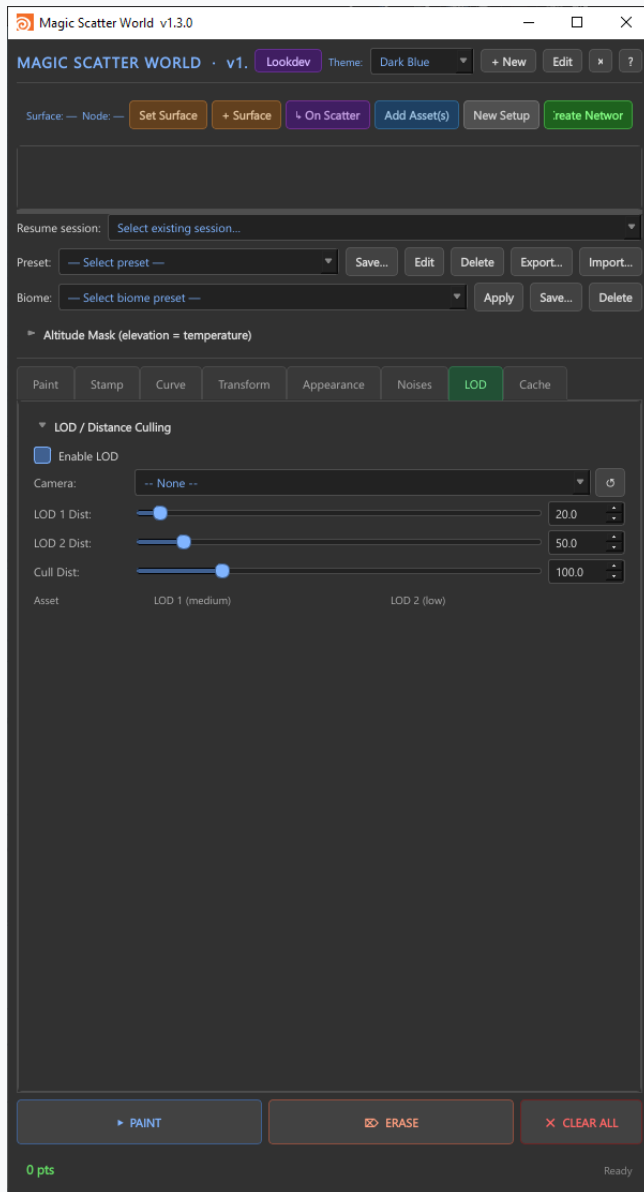
Noise Pattern — spatial frequency, offset, UV mapping

Animation — time-based noise evolution for growing scatter

Fractal — octaves, lacunarity, roughness for fine detail layering

Post-Process — remap and clamp noise output before it hits scatter

LOD Tab · Distance-based Level of Detail and instance culling



LOD / Distance Culling

Enable LOD — toggles the entire distance-based LOD system

Camera — the camera node used to measure instance distance

Distance Tiers

LOD 1 Dist — switch full-detail → medium asset (default: 20 units)

LOD 2 Dist — switch medium → low-detail asset (default: 50 units)

Cull Dist — remove instances entirely beyond this distance (default: 100 units)

Setup

Add three variants per asset type: high / medium / low detail

LOD system maps them to the three distance tiers automatically

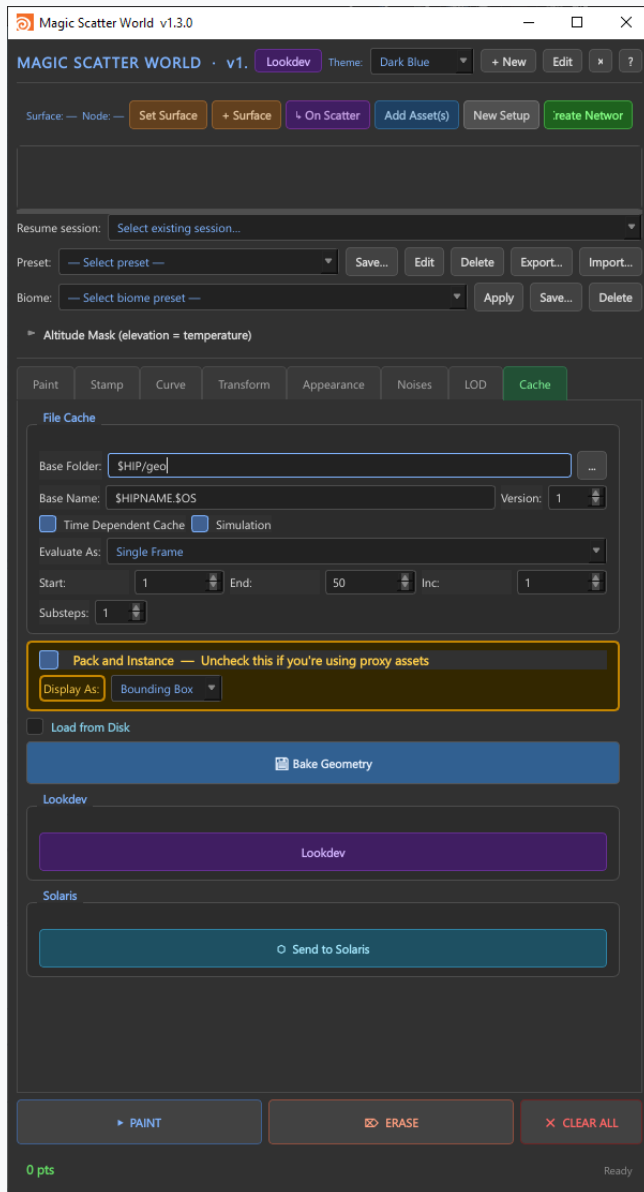
Combine with Camera Frustum Culling for maximum render performance

Workflow Tip

Start with Cull Dist at 200–500 units and tune inward

Verify tier transitions are invisible from all camera paths

Cache Tab · Bake to disk · Lookdev materials · USD / Solaris export



File Cache

Base Folder — output directory (\$HIP/geo writes beside the .hip file)

Base Name — filename template (\$HIPNAME.\$OS)

Version — increment to write new version without overwriting previous

Time Dependent Cache — one file per frame for animated scatter

Bake Geometry — freeze scatter geometry to disk

Pack and Instance

Uncheck when assets are Redshift Proxy (.rs) or Arnold Standin (.ass)

Checked = Houdini packed instancing for standard meshes

Unchecked = USD pipeline preserves proxy references end-to-end

Display As — Bounding Box (fast) or Full Geometry in viewport

Lookdev

Non-modal dialog: engine picker, 5 texture slots, 14 PBR parameters

Supports Arnold (HtoA) and Redshift native material networks

Karma/USD via MaterialX with UDIM and explicit UV readers

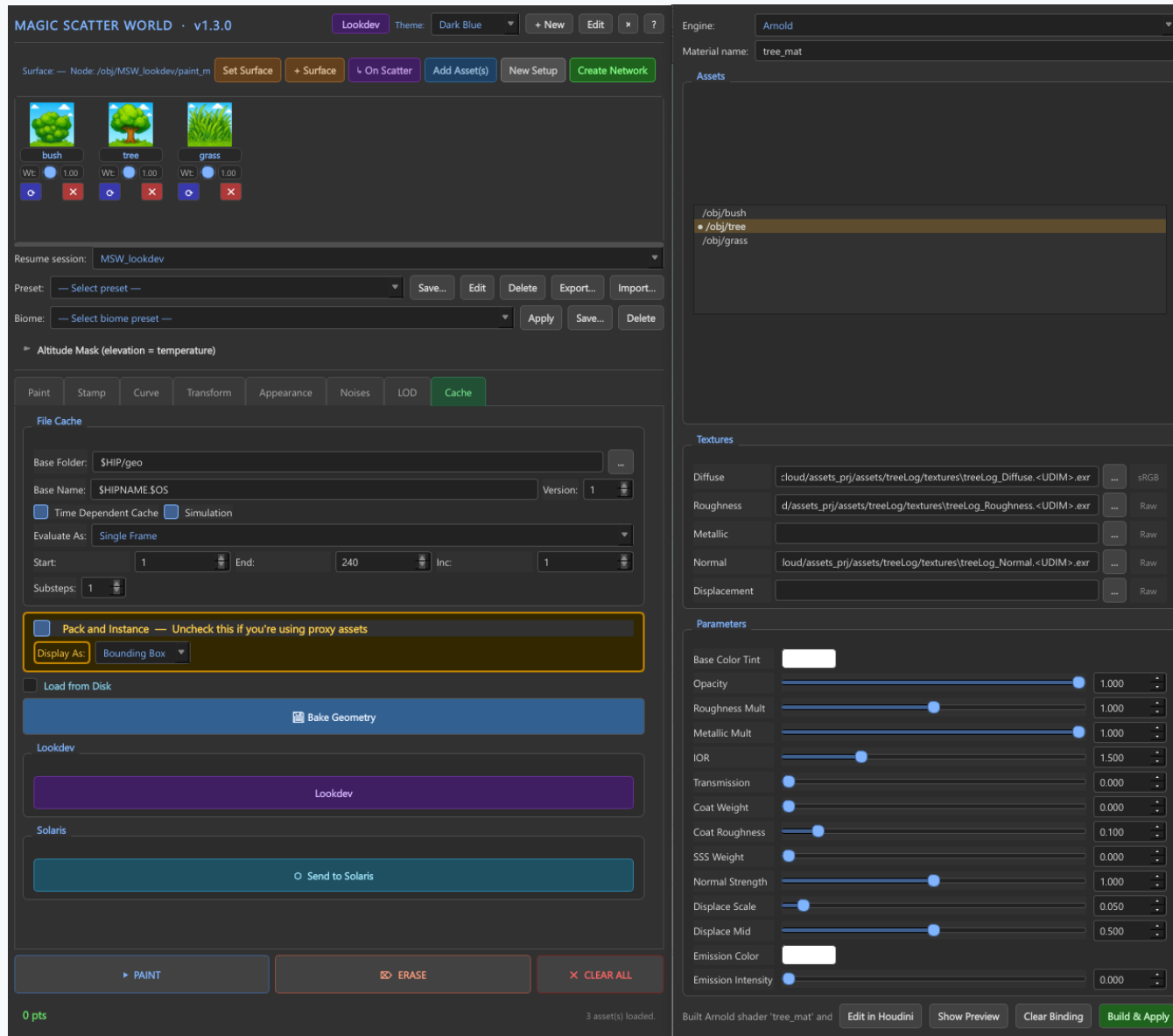
Send to Solaris

Exports as USD PointInstancer into a Solaris LOP network

Prototypes cached at \$HIP/usd/prototypes/ with SHA1 invalidation

Redshift Proxy and Arnold Standin assets preserved through export

Lookdev · PBR material assignment · Arnold · Redshift · Karma / USD



Engine Selection

Arnold (HtoA) and Redshift native material network builders
Auto-detects the available renderer on tool launch

Texture Slots

5 slots: Diffuse · Roughness · Metallic · Normal · Displacement
Browse for any file path — supports UDIM glob patterns
Per-engine colorspace badges (sRGB / Raw) per slot

PBR Parameters

Roughness Mult — scale the roughness texture output
Metallic Mult — scale the metallic texture output
Normal Strength — intensity of the normal map
Coat Weight / Coat Roughness — clearcoat layer controls
SSS Weight — subsurface scattering amount
Emission Intensity / Emission Color — self-illumination
IOR — index of refraction for specular response
Transmission — glass and water transparency

Actions

Build & Bake — compile material network and assign to all assets
Show Painting — toggle asset visibility in viewport
Bake Geometry — freeze geometry with material assignments to disk
Send to Solaris — export with MaterialX shaders for Karma / USD

Quick Start Workflow

· From zero to scatter in minutes

1. Open from the Magic Scatter World shelf button
2. Select your terrain → click Set Surface
3. Select asset geometry → click Add Asset(s)
4. Click Create Network and name your setup
5. Paint tab → adjust Radius and Density
6. Click PAINT and drag across the surface in the viewport
7. Use ERASE to clean up unwanted areas
8. Transform tab — tune rotation, scale, normal alignment
9. Appearance tab — add clumping, slope filter, color variation
10. Noises tab — add procedural density variation
11. Cache tab → click Bake Geometry to freeze the result
12. Click Send to Solaris for USD / Karma rendering

All settings are saved automatically inside the .hip file — no sidecar files required.